

April 27, 2026



*Advanced Molecular Detection
Southeast Region*

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UPDATES

- **In-Person Training (July 16–17, Athens, GA):**
 - Two-day, on-site workshop focused on hands-on genomic epidemiology tools, guided exercises, and real-world applications.
 - Supports preparation for the 12-week series through interactive sessions, pipeline walkthroughs, and small-group work.
 - **Please begin working** on internal **travel approvals** for 2 people, ideally one epidemiologist and one lead laboratorian
- **15-week training online – expansion on Genomic Epi**
- **HiPerGator**
 - Changes to how we request new accounts
 - Still figuring out the system and will update our directions as needed
 - IF you have any issues with accessing HPG, let Molly Mitchell know ASAP



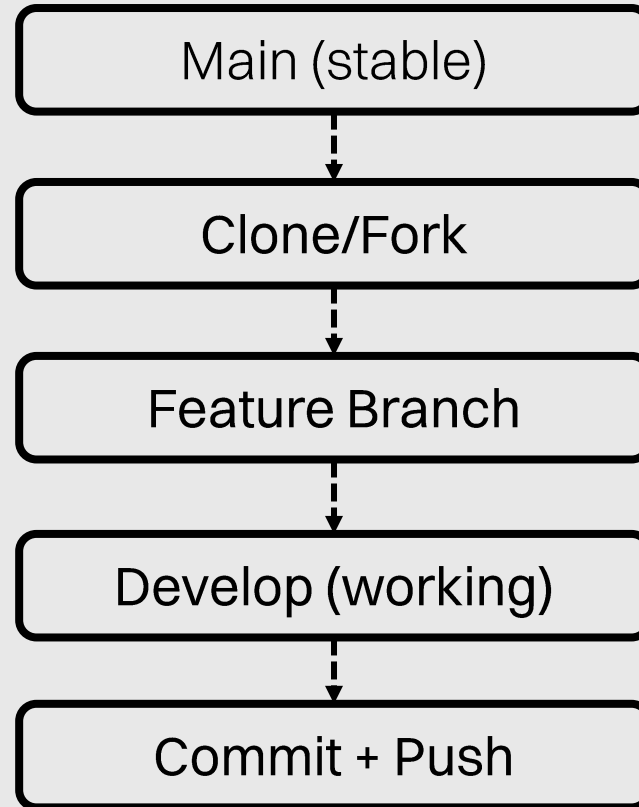
GitHub Vs. HPG

- GitHub → Code Management
- HPG → Pipeline Execution
- Together → Complete Workflow

Core Concepts

- › Repository = Project
- › Commit = Save
- › Branch = Separate Work
- › Pull Request = Review

Workflow



Review Process

- Create a pull request
- Assign reviewers
- Address feedback
- Merge after approval

Repository Structure

- **Core Files**

- > README.md
- > LICENSE
- > CHANGELOG.md
- > Nextflow.config
- > params.yaml
- > main.nf

Side Note:

- > Example for Nextflow Pipelines

- **Common Folders**

- > modules/
- > subworkflows/
- > bin/
- > assets/
- > test/

- **Environment**

- > environment.yaml

License

- Defines how code can be used and shared
- Required repository file
- MIT → default
- Other Options:
 - › Apache 2.0
 - › GPL v3
 - › CC BY 4.0
 - › Custom / Institutional

Versioning

Version Numbering Schema

X.Y.Z

- **X → Major Change**
- **Y → Minor Change**
- **Z → Patch Change**
- **Side Note:**
 - › Tag stable releases on main

Best Practices

- **Do not** work directly on main!
- **Do not** include sensitive data
- **Use** pull requests
- **Write** clear commit messages
- **Keep** documentation updated

Live Demo

1. Create/Clone Repository Locally
2. Create Develop Branch
3. Add/Change Key Files
4. Commit + Push → GitHub (Main Exist)
5. Open Pull Request
6. Review & Merge

Git Commands

One-time setup

- `git config --global user.name "Your Name"`
- `git config --global user.email@email.com`

SSH keys

GitHub removed password authentication for git in August 2021. Use SSH keys (recommended) or a Personal Access Token. We can provide guidance on this.

Creating & Cloning

- `git init` # start a new local repo
- `git clone <url>` # copy an existing repo to your machine

Pulling

- `git pull` # fetch latest changes/updates from remote
- `git fetch` # fetch without merging (check what changes/updates are out there)

Branching

- `git branch` # list local branches
- `git checkout -b <branch-name>` # create and switch to a new branch (use this one)
- `git checkout <branch-name>` # switch to an existing branch
- Never work directly on `main`. Always branch first.

Git Commands Cont.

Staging & Committing

git status # see what changed
git diff # see line-by-line changes before staging
git add <file> # stage a specific file
git add . # stage everything
git commit -m "your message" # commit with a short, descriptive message

Pushing

git push -u origin <branch-name> # first push of a new branch (sets upstream)
git push # subsequent pushes

Useful Extras

git log --oneline # condensed commit history
git restore <file> # discard local changes to a file

Resources

- **GitHub SOP - External (Available upon request)**
- **GitHub Pipeline Review Job Aid - External (Available upon request)**
- **GitHub Documentation**
- **Nextflow Documentation**
- **HPG to VS Code OH**
 - https://github.com/StaPH-B/southeast-region/blob/master/trainings/office%20hours/20250714_Session51_VScode%20Training%20Pt%201.pdf
- **Contact Us**
 - > **Request SOP:**
 - bphl-sebioinformatics@flhealth.gov



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

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